

A (Rather) Brief Reply to Pullum(2026)

Charles Han¹

Department of linguistics, Beijing Language and Culture University

Section 1 Prologue

I still have a vivid memory of me listening to Professor Geoffrey Pullum's lecture "Pirahã syntax, 'primitive' languages, and the Everett controversy²", in which he mentioned his friendship with Professor Steven Pinker when discussing the **Chomsky-Everett debate**. Although I cannot remember his words precisely, but the main point, if my memory is correct, was that **he and Professor Pinker, though disagree with each other sometimes, are still close friends**. Put the debate aside, I respect Professor Pullum and agree with him on that we shall agree to disagree.



Screenshot from my twitter

For readers who are interested in this debate, I recommend Pullum (2024) and Moore(2026), after all, listening to both sides makes one enlightened (兼听则明). While I choose **NOT** to get to the who's-right-who's-wrong business here, there is one thing I want to mention though, that is, the debate seems to be surrounding the observation that "Pirahã does not exhibit recursive structures in its syntax (taken from Sakel, Stapert & van der Hulst, 2010: introduction³ as an example)". If this is the case, then it shall be made clear that what's called recursion in generative linguistics represents **a computational mechanism** whose exact form shall be made clear by the inquiry, not merely certain type of structure. This interpretation of recursion could be

¹ contact information : hanganglbs@gmail.com

² <https://www.mpi.nl/events/mpi-colloquium-prof-dr-geoffrey-k-pullum>

³ https://daneverettbooks.com/wp-content/uploads/2023/11/Sakel_Stapert_recursion_revised.pdf

seen even from some early literatures of generative linguistics, for example, in Chomsky (1963: p.299): “The recursive property of the grammar can be attributed entirely to these transformations”. That the brain/mind is finite yet the number of potential sentences any individual (whose relevant mental mechanism is intact) could generate is infinite⁴ entails the existence of such a universal mechanism, whatever that mechanism is. Even if one takes the statement that “if there are recursive structures in a language L, then L’s syntax contains the recursive mechanism” to be true, the converse/inverse⁵ does NOT have to be true, this could be easily demonstrated by the following intuitive examples:

- 1) If John shoots some one, then he has a gun.
- 2) if John has a gun, he shoots someone.
- 3) if John doesn’t shoot someone, then he does not have a gun.

Loosely speaking, 1) is always true, but 2) & 3) could be either true or false.

Therefore, the observation that there is no such structure in Pirahã at most indicates that the precise definition of recursion shall subject to revision, neither that the notion recursion is not precisely defined previously, nor that this universal mechanism does not exist. While the former claim might result from the unfamiliarity with the history of generative grammar, the associated claim concerning UG is even a stranger position to take, as Chomsky (2000) wrote:

To say that language is not innate is to say that there is no difference between my granddaughter, a rock and a rabbit. In other words, if you take a rock, a rabbit and my granddaughter and put them in a community where people are talking English, they’ll all learn English. If people believe that, then they believe that language is not innate.

Section 2 Main Text

Now let’s take a look at Professor Pullum’s review of Postal(2025). I shall start from the biology section, and leaves the introduction section and first section aside.

⁴ Maybe unboudned is a better term, see Steedman(2024), but the term is used in a different context there.

⁵ I’ve noticed the earlier version of this draft contained a terminological mistake, the inverse of $P \rightarrow Q$ is $\neg P \rightarrow \neg Q$, the negation of $P \rightarrow Q$ is $P \rightarrow \neg Q$.

And I'll take Professor Pullum's interpretations of Professor Postal's criticisms to be totally faithful.

- [p.3] As early as 1969, Chomsky asserted that “**language has no objective existence apart from its mental representation**” (Chomsky 1969: 91, n. 3; quoted by Postal on p. 8 from a later reprinting). Postal regards this as extremely problematic.

Remark: The original text of Chomsky (1969:91, n.3) is:

Since **the** language has no objective existence apart from its mental representation, we need not distinguish between “system of beliefs” and “knowledge,” in this case.

Furthermore, the note is given within the context:

One might adopt the following research strategy for the study of cognitive processes in humans.

...

A perceptual model that relates stimulus and percept might incorporate a certain system of beliefs, certain strategies that are used in interpreting stimuli, and other factors...In the case of language, the technical term for the underlying system of beliefs is “grammar, or generative grammar.” A grammar is a system of rules that generates an infinite class of “potential percepts,” each with its phonetic, semantic, and syntactic aspects, **the class of structures that constitutes the language in question.**

...

To study it, we must abstract away from the other factors that are involved in the use and understanding of language, and concentrate on the knowledge of language [note 3] that has been internalized in some manner by the language user.

It seems to me the criticism arises from the failure to appreciate the difference between potentiality and actuality, a pair of concepts that could trace back to Aristotle⁶.

Here “the language” is explicitly defined as a collection of **potential percepts**, which in principle could be generated by some internal mechanism, the grammar, but are not there until they are generated. In other words, they have no objective existence until they are generated by the grammar. Such an interpretation, if not totally wrong, would indicate that the statement does not bear any incoherence in it.

- [p.4.] Postal makes some good points about such **pseudo-neurophysiological pretensions**. For example, he considers Chomsky's remark: “**So little is now known about the relevant aspects of the brain that we can barely even speculate about what the connections might be**” (Chomsky 1986: 39) and compares it with his statement the following year: “**I understand talk about the mind to be talk about the brain**

⁶ https://en.wikipedia.org/wiki/Potentiality_and_actuality

undertaken at a certain level of abstraction from (as yet unknown) mechanisms” (Chomsky 1987: 178). This simply does not make sense in terms of the usual notion of **scientific abstraction**. Scientists use abstractions to **reduce complexity of known physical phenomena**: treating a planet as a perfect sphere or as a point mass to make astronomical computations simpler, and so on.

Remark: Given a relatively complete picture, Chomsky(1986: 38-39) is in fact talking about the relationship between the theory of mind and the theory of brain:

“This enterprise is conducted at several levels: an abstract characterization in the theory of mind, and an inquiry into mechanisms in the brain sciences. In principle, **discoveries about the brain should influence the theory of mind**, and at the same time the abstract study of states of the language faculty should **formulate properties to be explained by the theory of the brain and is likely to be indispensable in the search for mechanisms**. To the extent that such connections can be established, the study of the mind-in particular, of I-language-will be assimilated to the mainstream of the natural sciences.”

So little is now known about the relevant aspects of the brain that we can barely even speculate about what the connections might be.

So the whole point is, at that time, so little is known about the connections between these two levels of researches such that we could only speculate about **the link** between them.

But the sentiment is obvious⁷, that is, the two lines of researches, exploring realities at different levels, shall inform each other in some principled way we might be able to formulate tentatively.

It is thus no wonder that in Chomsky(1987: 178), the author wrote:

The focus of discussion is knowledge of language: (1) what it is, (2) how it arises in the individual, (3) how it is put to use, and **(4) what brain mechanisms have the properties that we discover in this empirical investigation.**

Here I understand talk about the mind to be talk about the brain undertaken at a certain level of abstraction from (as yet unknown) mechanisms, at which we can formulate an explanatory theory that provides insight into the questions (1)-(3) and **serves as a guide for the inquiry (4).**

⁷ In fact, this systematic synthesis of biology and mathematics in the scientific study of language has already been deeply rooted in the generative tradition, as can be told from the works done by Professor Chomsky and his colleagues, Andrea Moro, Angela D. Friederici, Charles Yang, George A. Miller, Ian Tattersall, Jeffrey Watumull, Johan J. Bolhuis, Marcel P. Schützenberger, Marc D. Hauser, Matilde Marcolli, Robert C. Berwick, W. Tecumseh Fitch, to name a few. Thus, the statement in Pullum (2026) that “though the commitment to experimentation is strikingly absent” is NOT truth either.

In a sense, this research strategy corresponds well to Marr (1982/2011: p.25)'s three levels of analysis⁸, which basically says that at the computational level, which Marr himself called “the most abstract (p. 22)” level, one figures out what is being computed and why it computes this not that, and at the second level one constructs the algorithms and relevant representations for what's being computed (algorithmic and representational level) and at the third level one seeks how they're implemented in a physical device (implementational level). It's well-known that Professor Chomsky has always been focusing on the computational level, in other words, competence. And in fact, Marr himself wrote:

Chomsky's (1965) theory of transformational grammar is a true computational theory in the sense defined earlier. (Marr 1982/2011: p.28)

This indeed has nothing to do with the “scientific abstraction” in Pullum(2026)'s sense (nor in the sense of Postal, 2025: p.13-14), which, I reckon, refers to the Galilean Idealization in Weisberg's sense, that is, “the intentional introduction of distortion into scientific theories” in order to “make them computationally tractable” (Weisberg 2007). It is thus obvious that the criticism is totally beside the point..

[p.5.] “....that human languages are systems of “discrete infinity” and his idea that language is purely mental. If linguistic phenomena exist solely in the brain, a finite organ the size of a cauliflower, then, as Postal sees it, nothing about language can be infinite.....Postal sees equivocation: if a language is an infinite set of generated sentences, then no language is in anyone's brain, because brains are finite; but if we shift terminology radically as proposed in Chomsky (1986) and say that what we mean by a person's language is just the generative grammar inscribed in their brain (their “I language”, in Chomsky's perversely misleading terms), then no language is infinite. Chomsky can't have it both ways: either language isn't biological, or it isn't infinite.”

Remark: The criticism here shows some degree of confusion, which appears to result from a mixture of the objects of “infinite” and “biological”. Discrete infinity, is a property of the language faculty, and according to Chomsky(2005: p.11):

Any such system is based on a primitive operation that takes n objects already constructed, and constructs from them a new object....With Merge available, we instantly have an unbounded system of hierarchically structured expressions.

⁸ Though I reckon his analytic framework might be inspired by Chomsky's competence-performance distinction to some extent.

The infinity here refers to the potential unboundedness of the number of hierarchically structured expressions (which are not sentences in the traditional sense) that could be generated, while the language faculty is biological.

Similarly, in Postal (2025), the author constantly criticizes the statement that “NL is both biological and infinite” (p. 19-20, here NL refers to natural language, according to Postal 2025: p.1), yet this is never Professor Chomsky’s view, as has been explained, and whatever follows from this, thus is obviously irrelevant..

[p.6.] Chapter 3, the lengthiest in the book, is almost entirely devoted to making a single point. Following Postal, define the term “Z-language” to mean “recursively enumerable set of finite strings over a finite vocabulary”.... but he holds that human languages (specifically English) cannot be validly modeled as Z-languages. His main basis for saying this is that **English does not have a finite vocabulary**.... He claims that the set of English sentences is not even a recursively enumerable set....

Remark: I think there is only one thing to mention, that is, in Chomsky (1963), “A set is called recursively enumerable just in case there is a Turing machine that accepts all strings of this set and no others”. The notion is **NOT** defined over the size of vocabulary.

[p.8.] In Chapter 4, Postal addresses a truly absurd statement that Chomsky has made several times: that “in a certain, rather deep sense, there is only one human language” (see Postal’s p. 152). Chomsky makes this claim only when addressing general audiences....Postal marshals all sorts of evidence for the obvious claim that human languages are multitudinous and wildly diverse, and mentions that in one 2016 interview **Chomsky himself acknowledges the existence of “a very wide variety of typologically different languages”** (see p. 164). But it was hardly necessary. The question is not whether humans use languages exhibiting radically different types of syntactic and morphological structure, but **what on earth induced Chomsky to make the absurd rhetorical move of appearing to deny it.**

Remark: Postal(2025) and Pullum(2026) both cited the sentence “...in a certain, rather **deep sense**, there is only one human language” from Chomsky(1994), yet it’s unclear why it is so confusing to both.

Deep inside, we’re all just human, yet you see people with different faces, colors etc.. The statement that “Human languages, demonstrate a high degree of diversity on the one hand, and demonstrate a high degree of uniformity on the other hand” seems non-contradictory at least to me.

Section 3 Epilogue

As I was writing this conclusive remark, I was staring at the sentence in Pullum (2026) that “This book will not change the minds of any true Chomsky followers; they will find defenses”, a sense of sadness crossed my heart. As a guy who is interested in mathematical linguistics, I’ve read some early works done by both of them and was once attracted by their geniuses. But what’s happening here is upsetting. The assertion above is nothing but an overt refusal to any rational conversation. If one finds defenses, then he/she is a true follower of Chomsky, if no defense is given, then it would indicate the criticisms are valid. Yet, one would not need to be a true xxx follower to defend what is correct and clarify the misinterpreted, and agree-until-disagree does not make one a true “follower” of someone. It’s also, of course, totally unproblematic to have doubts and express them, but rational conversation shall avoid preconceptions, otherwise, the conversation is doomed to be going nowhere. Similarly, the field is going nowhere if linguists are busy dismissing each other.

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